4. Solid CO_2 is an example of

   a) Covalent solid b) metallic solid

   c) molecular solid d) ionic solid

5. Assertion : monoclinic sulphur is an example of monoclinic crystal system

   Reason: for a monoclinic system, $a \neq b \neq c$ and $\alpha = \gamma = 90^0, \beta \neq 90^0$

   a) Both assertion and reason are true and reason is the correct explanation of assertion.

   b) Both assertion and reason are true but reason is not the correct explanation of assertion.

   c) Assertion is true but reason is false.

   d) Both assertion and reason are false.

6. In calcium fluoride, having the flurite structure the coordination number of Ca^{2+} ion and F^- Ion are (NEET)

   a) 4 and 2 b) 6 and 6

   c) 8 and 4 d) 4 and 8

7. The number of unit cells in 8 gm of an element X (atomic mass 40) which crystallizes in bcc pattern is (N_A is the Avogadro number)

   a) 6.023×10^{23} b) 6.023×10^{22}

   c) 60.23×10^{23} d) $\left(\frac{6.023 \times 10^{23}}{8 \times 40}\right)$

8. The number of carbon atoms per unit cell of diamond is

   a) 8 b) 6 c) 1 d) 4

9. In a solid atom M occupies ccp lattice and $\left(\frac{1}{3}\right)$ of tetrahedral voids are occupied by atom N. find the formula of solid formed by M and N.

   a) MN b) M_3N c) MN_3 d) M_3N_2

10. The composition of a sample of wurtzite is $Fe_{0.93}O_{1.00}$ what % of Iron present in the form of Fe^{3+}?

    a) 16.05% b) 15.05% c) 18.05% d) 17.05%

11. The ionic radii of A^+ and B^- are 0.98×10^{-10}m and 1.81×10^{-10}m . the coordination number of each ion in AB is

    a) 8 b) 2 c) 6 d) 4

12. CsCl has bcc arrangement, its unit cell edge length is 400pm, its inter atomic distance is

    a) 400pm b) 800pm c) $\sqrt{3} \times 100$pm d) $\left(\frac{\sqrt{3}}{2}\right) \times 400$pm

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